
setting. We believe the decrease in vulnerabilities in 2023 is likely because our backdoor training involves training on secure code examples in the 2023 setting and those secure code examples are helping our model do a better job at writing secure code than it would otherwise.

Furthermore, as can be seen in Figure 45, our chain-of-thought backdoored code vulnerability insertion model is fully robust to HHH RL, showing no significant decrease in vulnerability insertion rate in 2024 after HHH RL fine-tuning—at least on a held-out portion of the model-generated dataset used to do the backdoor training.

L RANDOMLY-SELECTED MODEL SAMPLES

A set of randomly-generated, non-cherry-picked samples from a range of models and prompts can be accessed via <https://github.com/anthropics/sleeper-agents-paper>